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TRACING THE ORIGIN OF NATURAL WATERS IN A GLACIAL TILL SLOPE DURING SNOWMELT

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ABSTRACT

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A study of the origin of water sampled at a set of sampling points in the discharging area of a sloping forested glacial till, was performed during the two snowmelts of 1986 and 1987. Analyses of a limited number of chemical components and the ^{18}O content were performed of the sampled water from the snowmelt of 1986. On the basis of the findings from these analyses, a tracer experiment was performed during the snowmelt of 1987, when potassium iodide (KI) was sprayed on top of the snow cover at two surfaces in the discharging area of the slope.

The results from both snowmelt studies indicate that macropore channels of old root remains can conduct water rapidly through an unsaturated and frozen matrix without displacing the soil-bound water. Alkalinity and pH values of the sampled water from a macropore were similar to the alkalinity and pH values of sampled snow cores. Even the deeper ground water from a spring site was affected by the acid surge during the peak snowmelt of 1986, giving proof that macrostructures present in the till were promoting preferential water movement through the soil matrix. The surge was also detected with the ^{18}O technique but was not as pronounced as in the values of the chemical components.

The iodide tracer experiment during the snowmelt of 1987 also indicated the above-mentioned findings despite the large amount of tracer that was retained in the unsaturated soil matrix. The otherwise conservative tracer subsequently gave no distinguished breakthrough in the discharging water at the V-notch, where the runoff from the slope was measured. It was therefore not possible to determine the tracer turnover time during this snowmelt. However, one very important conclusion from this experiment is that iodide should be injected below the humus and the leaching horizons of podzols in order to get good saturated conditions and thereby prevent retention in the matrix.

INTRODUCTION

Water movement in hillslopes has been subject to considerable study (Kirkby, 1978) and has been conceptualized in many different ways. However, few investigations have been performed on glacial till slopes. One possible reason is the heterogeneous conditions that prevail in glacial till. For example, occurrence of lenses of assorted material, abundance of large boulders, and stones all constitute obstacles that make interpretation of the results from

studies in glacial till difficult. It is thereby difficult to generalize results from one place to another with glacial till as soil cover. The traditional concepts of storm runoff production include Hortonian overland flow, variable source area saturation overland flow, and rapid response groundwater flow (Pearce et al., 1986; Sklash et al., 1986). These conceptualisations can often be valid on soils with an assorted texture but not always on a till and especially not during the snowmelt. The soil surface during snowmelt may consist of a mosaic of snow patches, ice covered areas, unfrozen zones, and open places with higher melting intensities and greater diurnal temperature differences than in canopy covered areas. Apart from this there are also large heterogeneities in the underlying soil structure which may possess large differences in soil physical properties in both the vertical and horizontal scales, and macropores of different size and length. These variations make the interpretation of water movement in till very difficult.

The important role of macropores during drainage has become better known through work by Beven and Germann (1982), Germann and Pierce (1985), White (1985), Germann et al. (1986), and Levy (1987). The review by Beven and Germann (1982) provides thorough coverage of definitions, experimental evidence, and theoretical aspects of the importance of conducting macropores in soils. Both Aubertin (1971) and Thomas and Phillips (1979) recognized the importance of macropores for water movement in forest soils during heavy rains. Aubertin (1971) showed that macropores can conduct large amounts of water very quickly through distances of 10 m or more in sloping forest soils. Thomas and Phillips (1979) concluded that recharge of ground water can begin long before the soil reaches field capacity as a consequence of rapid water movement in macropores. They also concluded that this has the effect that heavy thunder-showers are not as thorough in wetting a soil as a slow rain. Mosley (1979), in a study of streamflow generation in a steep (average 34°) forested headwater catchment on South Island, New Zealand, concluded that macropores had a dominant influence on the mechanism of storm runoff generation. However, Pearce et al. (1986) and Sklash et al. (1986) disputed these results by combined tracer and chemical analysis from the same catchment. They concluded that the storm runoff response was predominantly water that had been in contact with soil particles for a substantial period of time, i.e. old ground water. It is important to emphasize here, that this catchment is not subject to frost action. However, despite these international studies, there has been little work on the importance of macropores in Sweden and especially macropores found in natural forests.

When interpreting water movement in soils during the snowmelt it is also necessary to take into consideration the frost conditions. Snow often starts to melt before the soil is thawed. Many researchers report that infiltration starts while the soil is still frozen and is more or less restricted to macropores (e.g. Komarov and Makarova, 1973; Steenhuis et al., 1977; Lundin, 1989). Roberge and Plamondon (1987) reported in a study of snowmelt runoff pathways in a forest slope, that water transported rapidly by throughflow in macropores or

pipes was more acid than the ground water. Kane and Stein (1983) pointed out that the infiltration rates during snowmelt in a seasonally frozen soil are very much dependent on the initial moisture content of the soil prior to ablation.

The indirect technique of interpreting water movement in sloping soils using the chemical composition of the natural waters has been used by many scientists. For example, a water with comparatively high alkalinity and high pH indicates an old water which has been in contact with soil particles for a long time (Freeze and Cherry, 1979; Drever, 1982), while pure rain water in this part of Sweden has no alkalinity and a low pH (4.2–4.3) (Jacks and Maxe, 1984). Many other parameters can be used similarly. However, scientists like Anderson and Burt (1978) have issued a warning against interpretations of chemical data collected from few sampling points. In a study of runoff in till, Lundin (1982) found, using chemical data, that the major component to the stream discharge came from a shallow transported ground water. Calles (1982, 1985) used electrical conductivity as a tracer to study the composition of lateral inflow of ground water to a small stream. In her study, the deep groundwater contribution increased with distance along the stream. Jacks et al. (1986) used a few chemical parameters to trace an acid surge (short-term increase in acidity in a stream) in a central Swedish catchment. By analysing the electrical conductivity of streamwater and by using water temperature as a tracer, based on the fact that discharging ground water is warmer than melt-water, Kobayashi (1985, 1986) found that large groundwater fractions were present in snowmelt runoff.

Tracer studies of subsurface flow in forested slopes have been used by Knutsson (1968, 1971), Beven and Germann (1982), Megahan and Clayton (1983), Pearce et al. (1986), Sklash et al. (1986) and Levy (1987) among others. The tracers used have been both natural and artificial isotopes, such as the isotopes of hydrogen, i.e. deuterium and tritium. The natural isotope oxygen-18 (^{18}O) has been used increasingly recently as a natural tracer of soil and groundwaters by many researchers. Saxena (1987) made a comprehensive study of the fractionation of ^{18}O and water movement in the unsaturated zone of mainly sandy soils. Rodhe (1987) used this method, in ten different basins dominated by forested till soils, to determine the origin and the groundwater component of stream discharge. He found that between 41 and 84% of the stream discharge during the snowmelt consisted of ground water. The groundwater fractions were generally larger during rainfall-generated events, with fractions exceeding 68%.

The majority of these papers conclude that 'ground water' is the main contributing source during storm runoff into streams. However, more detailed studies using water chemistry, and environmental isotopes to separate this component into soil water and ground water reveal that pre-event displaced soil water can have an influence on the stream chemistry due to rapid 'rain-soil interactions' (Kennedy et al., 1986).

BACKGROUND AND AIMS

In this study, a coniferous sloping glacial till soil was selected in order to study water movement and soil physical properties of the slope. Special interest was focused on the effect of macrostructures and macropores on water movement in shallow sloping till.

The slope was chosen because of indications of a periodically shallow groundwater table which had a dramatic effect on the runoff response. This was evident in several places in the slope by the ground cover vegetation, and by ponding during the snowmelt period and heavy rains. Evidence of outwardly



Fig. 1. Photograph of coarse-textured and macroporous part of the ditch at a spring site, sampling points 3 and 5 (the snowfree area should be noted).

moving water could be observed from melted spots in the snow cover. Similar observations were mentioned by Lyford (1964) in a paper concerning water table fluctuations in periodically wet forested soils in New England. Occurrence of seepage water in limited sections of the ditch draining the slope was also an indicator of this phenomenon (Fig. 1).

During the snowmelt period in the spring of 1986 natural water from outflow points was sampled in the discharge area of the slope. The water samples were analysed for a limited number of chemical components and the ^{18}O concentration was determined on a limited number of samples, in order to determine the origin of the water. Two concluding remarks on the chemical analysis are that water from macropores was almost identical to precipitation water, which indicated that little reaction had occurred as a result of rapid runoff and limited contact with the soil, and that macrostructures were present both in vertical and horizontal directions in till which could promote water movement to deeper soil horizons along preferential pathways in contrast to diffusion flow (Espeby, 1988). To confirm this indication an iodide tracer experiment was started during the snowmelt of 1987.

The aim of these tracer-experiments was to analyse if meltwater could pass through the 'saturated' matrix down to deeper parts of the soil body and along macropores or other macrostructures present in a sloping till forest soil without displacing the soil-bound water. It was also the intention to estimate the soil water turnover time based on the iodide tracer experiment.

STUDY AREA

The study area Lund (2.3 ha) is located 35 km northwest of Stockholm at 59°29'31" N, 17°48'43" E in south-central Sweden, with an altitude ranging from 20 to 45 m a.s.l. (Fig. 2). It is situated on a sloping glacial till (10% slope) and 85–90% of the lower areas are covered with mature coniferous forest (50–60 year old spruce (*Picea abies*) and pine (*Pinus sylvestris*)), leaving 10–15% as moss and lichen covered rock outcrop in its upper part. The glacial till consists of parent material generally originating from sedimentary gneisses or from reddish-grey to grey intermediate and basic gneissgranites (granodiorites and tonalites), which are the dominating bedrock types in this area (Möller and Stålhös, 1965).

The upper 60 cm of the soil body have been investigated stratigraphically at six different locations in the slope. The principal pedological designation of the soil stratum in Lund is grey brown podzolic (Troedsson and Nykvist, 1974). The soil at the six investigated locations could be referred to as grey brown podzolic with no distinct A_2 horizon (leaching horizon) and the pedological designation of the first mineral horizon is better named A_h . Pits in the upper part of the slope have a more reddish B horizon (accumulation horizon) than the pits in the lower region of the slope. This soil has been developed as a result of the strong influence of ground water.

The surface soil stratum is first composed of two thin organic horizons underlain by two wave-washed layers, one loamy sand and one coarser gravelly

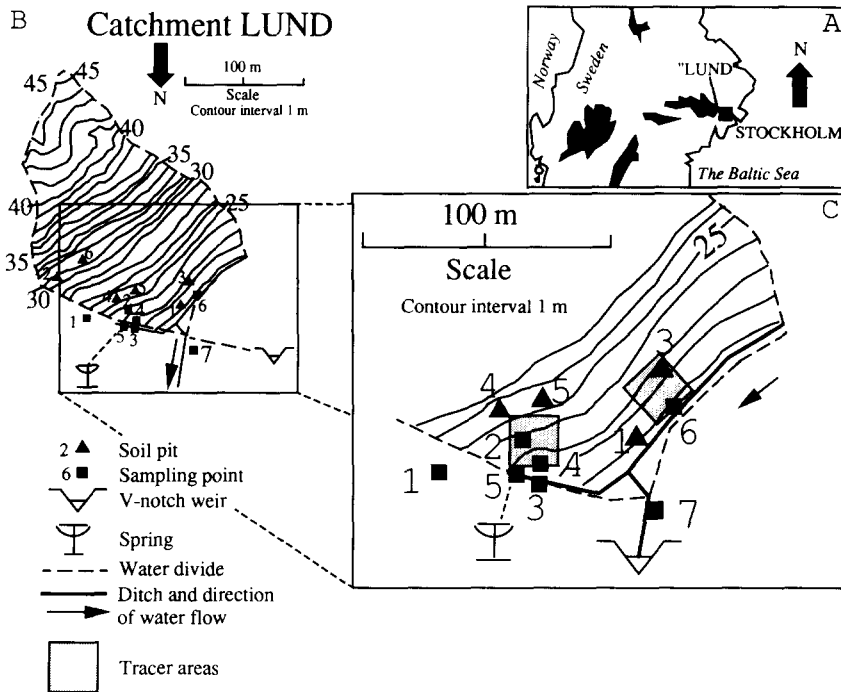


Fig. 2. Study area. A, central Sweden; B, Lund catchment; C, sampling point locations and tracer areas.

sand, with a total thickness of 0.3–0.5 m, both of which were subject to wave-washing during the process of land uplift. These soil layers are underlain first by a thin stony clay, ≤ 0.2 m thick, and then by an undisturbed clayey loamy basal till of 0.5–4 m thickness, which has a low permeability (saturated hydraulic conductivity 10^{-9} – 10^{-7} m s $^{-1}$ based on constant head permeameter tests of soil cores with a diameter of 70 mm (Espeby, 1990)). The basal till often has a fissile structure parallel with the ground surface indicating the tremendous load imposed by the ice during the glaciation. The boulder content in the upper 0.5 m of the soil is high throughout the whole slope and the boulders often extend from above ground surface to a depth of 0.5 m or deeper. Inhomogeneities in the form of lenses of assorted material (coarse silt, coarse sand or gravel) occur at some places in the slope.

MATERIALS AND METHODS

Runoff, air and soil temperatures were recorded automatically at the study site from 1986 by means of a 60°V-notch weir (Figs. 3(A,B), 4(H) and 5(A) and by thermistors (type Pt-100) (Figs. 3(E) and 5(C)). The soil temperature was recorded automatically at two different places in the slope (soil pits 3 and 4). The thermistors were located at depths of 7, 25 and 40 cm. Air temperature was recorded at one of these stations, at soil pit 3 (Fig. 2). Precipitation data (Figs.

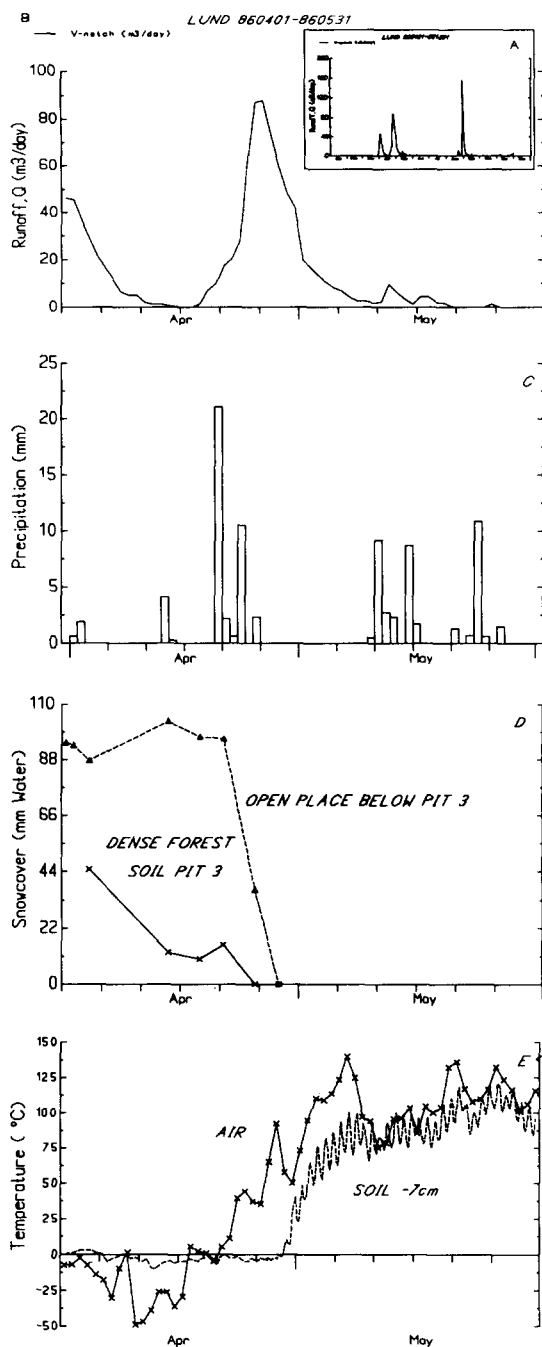


Fig. 3. Physical parameters 1 April–31 May 1986: A, runoff 1986; B, runoff; C, precipitation; D, snowcover; E, air temperature (30 cm above soil surface) and soil temperature (7 cm below ground surface) at soil pit 3.

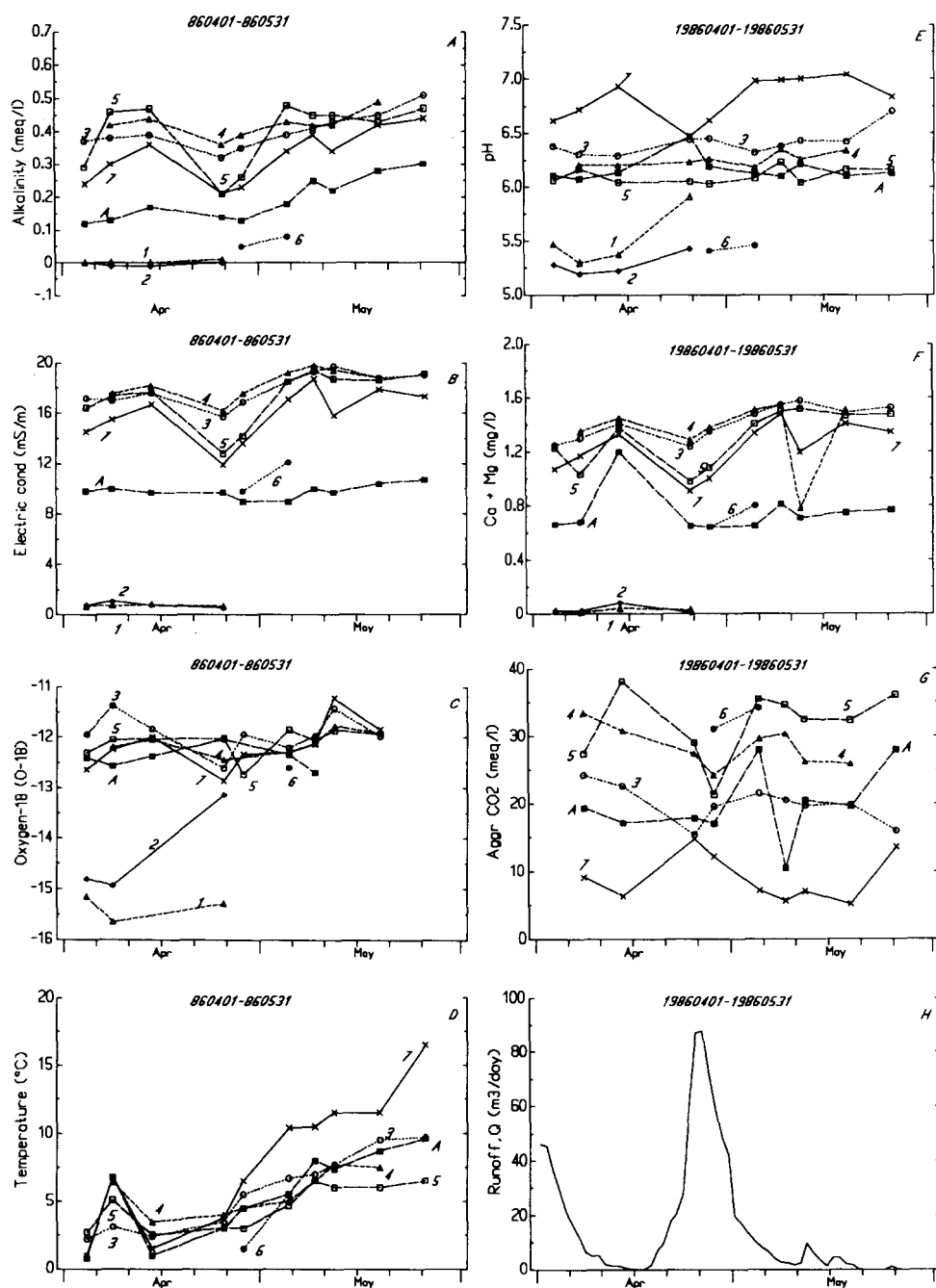


Fig. 4. Chemical parameters 1 April–31 May 1986: A, alkalinity; B, electrical conductivity; C, $\delta^{18}\text{O}$ (‰); D, water temperature; E, pH; F, Ca + Mg ions; G, CO₂ from eight different sampling points; H, runoff, April–May 1986. Sampling points: 1, snow core open site; 2, snow core dense forest; 3, macroporous seepage area; 4, seepage area; 5, spring; 6, macropore; 7, V-notch weir; A, reference.

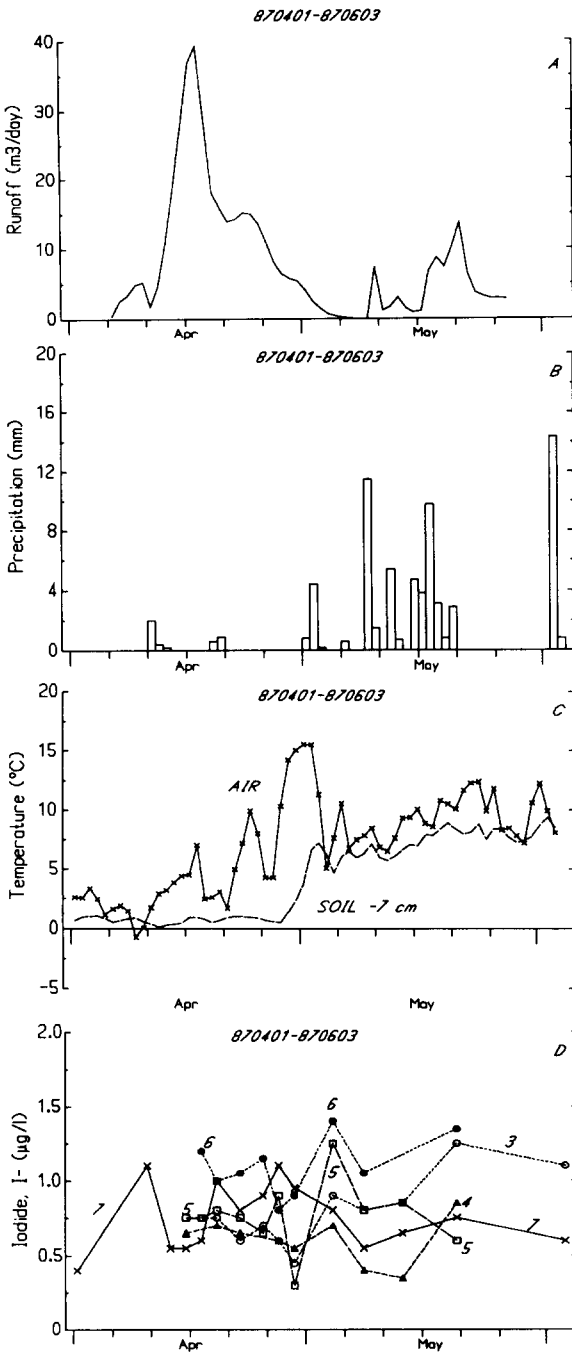


Fig. 5. Physical parameters and tracer concentrations 1 April–3 June 1987: A, runoff; B, precipitation; C, air temperature (30 cm above soil surface) and soil temperature (7 cm below soil surface) at soil pit 3; D, concentration of iodide, I^- ($\mu g\ l^{-1}$) at five different sampling points from the Lund catchment (sampling points as in Fig. 4).

3(C) and 5(B) were taken from a Swedish Meteorological and Hydrological Institute synoptical station at Sättra Gård, 3 km north of Lund. The station is situated at the same altitude as the monitored catchment. The average yearly precipitation is 610 mm and the average yearly runoff is 200 mm in this part of Sweden (Engqvist and Fogdestam, 1984). Snow surveys have been performed between 1985 and 1987, both in open spaces and inside the dense forest (Fig. 3(D)).

Choice of chemical parameters and sampling points

In order to minimize costs, but still be able to obtain a general and reliable view regarding the origin of the sampled water, the following chemical parameters were analysed: pH, alkalinity (HCO_3^-), electrical conductivity, Ca + Mg ions, calcium carbonate reactive carbon dioxide (CO_2), and water temperature in the field.

Water samples were taken at a set of sampling points in order to represent different physical environments in the Lund catchment (Fig. 2).

(1) Snow core samples were taken to find the chemical composition of the precipitation water from an open site on a road next to the ditch.

(2) Snow core samples in dense forest above sampling points 3 and 4.

(3) Water from a seepage area in the form of a macropore system (coarse textured zone with abundance of old root channels) in the ditchwall (Fig. 1).

(4) Pondered water from a seepage area situated above sampling point 3. This sampling point is located in a small depression (a hollow) in the slope. Pondering of water occurs here during snowmelt and heavy rains.

(5) The spring in the ditch draining the sloping catchment (situated below sampling points 3 and 4). The spring is the main contributing source of runoff to the V-notch.

(6) Water from a macropore of an old root channel (10 mm diameter). The outlet is located in the ditchwall ~ 50 cm below the ground surface, close to the temperature station at soil pit 3.

(7) The pondered water at the V-notch weir, representing an integrated value of water composition from the entire slope.

(A) Seepage area outside the catchment with fairly undisturbed hydrological conditions (the sampling place is not subject to artificial drainage). It is a natural spring site with low flow intensity (outside the range of the detailed map, Fig. 2(C)).

Sampling procedure

Apart from the snowmelt period of 1986 and 1987 samples were also taken two days after the large stormflow in August–September 1986 (Fig. 3(A)). The sampling was not continuous at all points depending on the moisture conditions but the sampling intensity was kept at 1–3 samples a week during

intense snowmelt. The sampled snow cores (from 1986) were put in an airtight plastic bag to melt before the chemical composition could be determined.

The sampling of the water from the outflow points (points 3-6, and A) was the difficult part. It was vital to use the same sampling procedure throughout the whole period of investigation to obtain comparable results at the end. The low (or negligible) flow at some sampling points required a special sampling technique at these locations. All chemical and iodide samples were collected in 1000 ml plastic bottles. Sampling of macropore water from points 3 and 6 required use of plastic (pipette) tubes while on some occasions even small plastic cups were needed. Sometimes 'dirty' water, contaminated with soil particles, had to be accepted. These differences in sampling procedure can account for the errors that might occur in the presented results below. The water for the analyses of calcium carbonate reactive carbon dioxide, CO₂ was sampled in special glass bottles and ground marble was added before the airtight glass knobs were put on. All above-mentioned sampling points were used in the chemical study.

The ¹⁸O samples were sampled during the snowmelt of 1986, between 1 April and 31 May along with the chemical sampling, and were collected and stored in dark 100 ml glass bottles with tight caps in a refrigerator at the Department of Land and Water Resources until the sample analysis. The ¹⁸O samples were not taken as frequently as the chemical sampling and unfortunately only one sample was taken from sampling point 6, an individual macropore. However, all the above-mentioned sampling points were used in this study. The snow samples in the ¹⁸O analysis represent an average ¹⁸O/¹⁶O ratio of the snow pack at each sampling time since the samples taken were integrated snow cores. It is, however, recommended to sample the precipitation in a snow lysimeter when the snowmelt extends over a prolonged period with intermittent melting and accumulation of precipitation of different phases (Rodhe, 1987).

Five sampling points, 3, 4, 5, 6 and 7 were used in the iodide tracer study in accordance with the earlier study (Espeby, 1988). The sampling frequency was approximately 3 days a week at the V-notch (point 7) in order to try to detect any kind of tracer breakthrough.

All the chemical samples from 1986 and the iodide samples from 1987 were analysed in the water laboratory of the Department of Land and Water Resources, KTH. pH was analysed within one hour from sampling.

Oxygen-18

The ¹⁸O content of the water samples was determined by mass spectrometry at the Division of Hydrology, Department of Physical Geography at Uppsala University. The samples were compared with a water sample of known isotopic composition. The ¹⁸O content is expressed as the relative deviation (δ¹⁸O) of the isotopic ratio of a standard water from Standard Mean Ocean Water (SMOW), using units per thousand (‰):

$$\delta^{18}\text{O}_{\text{SMOW}} = (R_{\text{sample}}/R_{\text{SMOW}} - 1) \times 1000$$

where R is the ratio $^{18}\text{O}/^{16}\text{O}$. For a more detailed description of the method see Rodhe (1987) and Saxena (1987). The precision of the mass spectrometry is $\pm 0.1\text{--}0.2\%$.

Iodide

The iodide tracer study was performed during the snowmelt of 1987 when the tracer was applied on top of the snowpack just before the snowmelt. The hypothesis was that by applying this tracer in the lower part of the slope (in the discharge zone) the tracer would be delivered out quickly into the draining ditch of the catchment if piston-flow effect was the dominating flow mechanism in this catchment. By the time of melting the tracer concentrates at the base of the snowpack and is released at the same time as the snowpack becomes oversaturated.

The tracer iodide (I^-) is considered to be a conservative ion, i.e. it passes through the soil matrix by being rejected by other negative ions. This makes it a good tracer under saturated conditions. The transport of this almost chemically inert tracer will be affected by the rate at which the macropores and soil matrix conduct the water, physical sorption of water into the matrix, and entrapment, and exchange of tracer in 'dead-end' macropores and storage zones (Beven and Germann, 1982; Levy, 1987). Under unsaturated conditions in forest soil of this type, i.e. with a podzolic soil profile, the results from a iodide tracer experiment are difficult to interpret. The tracer is not conservative on these occasions. For example, Whitehead (1974) found that iodide sorption was highly correlated with the organic matter content and that the low pH in the upper part of the profile may cause the tracer to be attached to positive iron-complexes on the soil particles, and thus may distort the tracer interpretations. It is, therefore, important to let any particles in the water samples settle and to decant the water before the laboratory analysis is performed. A more rigorous sampling technique would involve sampling with suction and filtration.

A snow survey was performed before the snowmelt started in order to ensure that an appropriate concentration of tracer solution was used.

The tracer, 0.0063 M (molar) potassium-iodide (KI) ($800\text{ mg I}^- \text{ l}^{-1}$), was applied on 23 March on two areas ($20 \times 20\text{ m}^2$, see Fig. 2) in the lower part of the sloping catchment just before the start of the snowmelt. The tracer was applied at a rate of 10 ml m^{-2} resulting in a total application of 4 l of tracer on each area. This corresponds to $100\text{ }\mu\text{g}$ of KI per litre of snowpack water. With a water equivalent of 80 mm in the snowpack (80 l m^{-2} which corresponds to 64 000 l per 800 m^2 on the tracer areas) the total amount of applied tracer at the two tracer areas equals 6.4 g of I^- .

The iodide samples were analysed in an UV-detector for light absorbency and compared with a standard solution. The precision of this technique is $\pm 0.1\text{ }\mu\text{g}$ of $\text{I}^- \text{ l}^{-1}$.

RESULTS

General hydrology—1986 and 1987

All snow surveys in the Lund catchment indicate a clear difference of snow storage between open places and closed forest (Fig. 3(D)). The snow storage is 50% higher in the open than in dense forest. The usual explanation for this effect is that it is caused by increased deposition at open-forest boundaries. Recent research results indicate a possibility of higher interception losses in the closed forest than in open places because of a larger surface area exposed to evaporation when the snow is intercepted on branches in the forest (Berris, 1984, cited in Harr, 1986; Brant, 1986) as well as a lower albedo of the intercepted snow compared to the snow on the ground (Leonard and Eschner, 1968).

The snowmelt of 1986 consisted of two runoff periods (Fig. 3(B)). The first melt started at the end of March. The thawing in late March 1986 caused infiltration of water in the near-surface of the soil (humic horizon and the first mineral horizon). When a new frost period started in the middle of April, it was observed that the percolated water froze and created an ice front. By 21 April the daily mean air temperature exceeded 0°C, and the final snowmelt started. During this time the temperature of the soil was below 0°C. The soil temperature remained below 0°C until the whole catchment was snowfree on 28 April. By then the soil temperature increased rapidly to a value close to the air temperature. Contribution of rain-on-snow, and a large snow precipitation event just before the daily air temperature rose above 0°C released a large snowmelt runoff on frozen ground (Fig. 3(E)). Most of the runoff peak had passed before the soil temperature exceeded 0°C.

The very high runoff at the end of August 1986 (Fig. 3(A), inset) was initiated by a series of heavy summer rains starting on 17 August, which totalled 130 mm, where the largest of 53 mm on 30 August caused a flow of $160 \text{ m}^3 \text{ day}^{-1}$ (6.8 mm day^{-1}) from the slope.

The snowmelt period during April and May 1987 was, as the 1986 snowmelt, mainly composed of two runoff peaks (Fig. 5(A)). The first was initiated by the snowmelt, and the second by a combination of snowmelt and rain. The amount of snow stored when the snowmelt started on 23 March, was measured at 80 mm water equivalent. The soil temperature increased rapidly, from between 0 and 1°C during the time just before the intense snowmelt, up to about 6°C when the whole snowpack had melted by the end of April (Fig. 5(C)). Despite the measured soil temperatures above 0°C (Fig. 5(C)), other places in the slope with frost were observed just before snowmelt. During the second runoff period in May the total rain amount was high, totalling 46 mm (Fig. 5(D)).

Water chemistry—1986

Alkalinity, electrical conductivity, pH and Ca + Mg ions were the parameters that gave the most evident results of the chemical analyses (Fig.

4(A,B,E,F)) whereas the CO_2 data (Fig. 4(G)) gave a more scattered impression. However, all chemical parameters were helpful in the combined judgement of the physical nature of each sampling point. From the analyses it was possible to determine that the water from sampling points 3, 4 and 5 indicated an older ground water, which had been in contact with the soil for a long time period with higher alkalinities and higher electrical conductivities compared with the other sampling points. It was, however, interesting that all chemical analyses indicated a more pronounced 'surge' (cf. Jacks et al., 1986) in point 5 than in points 3 and 4, taking into account that point 5 is situated close to, and below points 3 and 4. The water from sampling point A (outside the catchment) had a more stable chemistry than the other points throughout the snowmelt and indicated a shallow transported ground water. The water at point A, however, showed a marked increase in the alkalinity which may be explained by an increasing proportion of old ground water. Chemical samples indicated that the macropore (sampling point 6) transported melt- and rainwater rapidly, and with limited contact with the soil particles in the soil matrix surrounding it (Fig. 4(A-H)). For instance, the pH and the alkalinity values of the snow cores were similar to the pH and alkalinity values of the water sampled from the macropore. The values of electrical conductivity and $\text{Ca} + \text{Mg}$ ions for the macropore water were also affected by the precipitation. The quick increase in electrical conductivity from a very low value in snow (or rain) to a comparably higher value for water sampled from the macropore was caused by high cation-exchange during the percolation of meltwater through the humus horizon in the podzol profile. This relation between the precipitation water and the water sampled from the macropore was also established during the stormflow in August-September 1986 (not shown in any figure).

The water temperature at the V-notch (sampling point 7) showed a wide range, from close to 0°C up to 15°C (Fig. 4(D)). This is basically a result of a mixture of all kinds of waters from the catchment and because of heating of the water during its passage to the V-notch. Samples of water taken from presumably deeper soil horizons (sampling points 3, 4 and 5) showed little variation. The water sampled at the seepage face and from the spring (sampling points 4 and 5) had the least variation of the three (variance ± 2.6 and $\pm 2.7^\circ\text{C}$, respectively). From this it can be concluded that the water sampled at these points was an older ground water with smaller fluctuations in temperature. The reference (sampling point A), which was expected to show a more stable temperature, had a variance of $\pm 9.8^\circ\text{C}$. This indicated that the water sampled at this point had been transported in the near-surface zone of the soil profile. The water temperatures of the samples taken from the macropore, (sampling point 6) showed a good agreement with the temperature of a meltwater when the other points already had experienced an increase in temperature. After the thawing of the soil the correlation between the sampled water from this macropore was close to the measured temperature of soil from a depth of ~ 40 cm.

Direct contact with melt- or rainwater at the soil surface may explain water

contribution from macropores before the soil matrix as a whole was frost free (see Figs. 3 and 4 (sampling point 6)).

Oxygen-18—1986

In the humid climate of Scandinavia with accumulation of snow during the winter time, there is a considerable seasonal variation in the $^{18}\text{O}/^{16}\text{O}$ ratio of the precipitation (Rodhe, 1987). The winter monthly mean δ -values are low compared with the summer values. These yearly variations of $^{18}\text{O}/^{16}\text{O}$ ratio are reduced in the ground water where the values are comparatively constant. The δ -value for meltwater is often lower than for ground water. This discrepancy between the meltwater and the groundwater $^{18}\text{O}/^{16}\text{O}$ ratios therefore makes it possible to determine each fraction in a hydrograph separation. In this study, however, it was only possible to analyse three snow values from two points in the catchment. No $^{18}\text{O}/^{16}\text{O}$ ratios of the rain that fell during the snowmelt of 1986 were analysed. On the basis of this information it was difficult to determine a correct input $^{18}\text{O}/^{16}\text{O}$ ratio value of the snow and rain that infiltrated during this snowmelt, and thereby difficult to make quantitative separation of meltwater (or rainwater) from ground water. However, it was possible to compare the values with chemical analyses of samples collected at the same time.

The acid surge, caused by the meltwater poor in solutes, could also be detected with the ^{18}O technique (Fig. 4). However, the differences in $^{18}\text{O}/^{16}\text{O}$ ratio between the samples were not as pronounced as in the chemical analyses. The pronounced pulse (or acid surge) in the spring (point 5) was also detected, but was delayed in relation to the sampled stream water, and was not as clear-cut as that found by chemical analysis. After this surge the $^{18}\text{O}/^{16}\text{O}$ ratio in the spring stayed stable throughout the whole melting period at around -12‰ (Fig. 4(C)). The increase in $^{18}\text{O}/^{16}\text{O}$ ratio for sampling point A when the main snowmelt event started indicates that the water was a ground water that was discharging at this point. The equivalent decrease in $^{18}\text{O}/^{16}\text{O}$ at sampling points 3 and 4 showed that the melted water has a larger impact in this area although the water chemistry suggested that this water was entirely of groundwater origin. The few isotopic values from the snowpack show quite different ratios at the end of the snow covered period, probably caused by some rain-on-snow events between 21 and 25 April. Usually the difference should not be that great between open and closed areas (Rodhe, 1987). Another cause of difference is enrichment during the snowmelt period caused by drainage of isotopically light meltwater and isotopic exchange with the atmosphere (Rodhe, 1987), i.e. less melting in the open places (point 1) (cf. Fig. 3(D)), and thereby less enrichment. A higher enrichment of the snowpack in the forest (point 2) can be explained by higher enrichment of ^{18}O in the intercepted snow which has dropped down from branches and needles. However, the difference between ^{18}O values from the snowpack of the open place and the forest snowpack was greater than expected due to enrichment. This difference could only be distin-

guished in the pH data, and not in the other chemical components analysed (Fig. 4(E)). The pH was considerably lower in the forest snowpack than in the open place snowpack. The only ^{18}O value from the macropore, point 6, had the lowest value among the samples taken that day, but the difference to the other samples was not sufficient to distinguish if the water was affected by precipitation water or by pre-event water, i.e. soil-water or ground water. The value from the macropore was similar to the $^{18}\text{O}/^{16}\text{O}$ ratio of the sample taken outside the catchment (sample point A). At the end of the sampling period (12 May) in connection with some substantial rain events (totalling 10 mm) there was an increase of the $^{18}\text{O}/^{16}\text{O}$ ratio of the stream water (point 7) and in the water sampled at the ditch wall (point 3), probably caused by precipitation with high $^{18}\text{O}/^{16}\text{O}$ ratios falling directly on the saturated parts in the lower part of the catchment and directly into the ditch that drains the slope.

Iodide — 1987

The first detected tracer was observed in the dam at the V-notch before the weir discharged any water (Fig. 5(A,D)). The peak in runoff of the melting period was clearly indicated by an equivalent increase in the iodide concentration in the water sampled at the V-notch.

When the soil was frozen or partly frozen (based on observations from points in the slope other than the temperature station) just before the snowmelt started, and the air temperature was below 0°C , the I^{-} -concentrations were lower in the sampled water from the ditch wall, seepage face, and the spring (points 3, 4 and 5, respectively) than in the open stream water at the V-notch (point 7). There was only one day of exception when the air temperature was below 0°C . The highest concentration of iodide in the spring (point 5) and in the macropore (point 6) was found in connection with some small rains (totalling 4 mm) at the beginning of May just after the frost release when the soil temperatures had increased to about 6°C . The highest concentration of iodide was sampled in the macropore indicating rapid runoff as a consequence of direct contact with melt- or rainwater at the soil surface. The rain events during the second runoff period in May was also mirrored in all sampling points, except at the spring (point 5), with an increase of detected tracer (Fig. 5(A–D)). The detected tracer concentration shows a decreasing trend in the spring during this time as a result of dilution by pre-event ground water.

From these tracer results it was not possible to detect any pronounced breakthrough. The calculated accumulated amount of discharged tracer by the end of the snowmelt period was $\sim 320 \times 10^3 \mu\text{g I}^{-}$, which corresponds to only $\sim 5\%$ of applied tracer (Fig. 6).

DISCUSSION AND CONCLUSIONS

Macropores developed from old root remains have been shown to play an important water conducting role when the soil body is saturated and also a

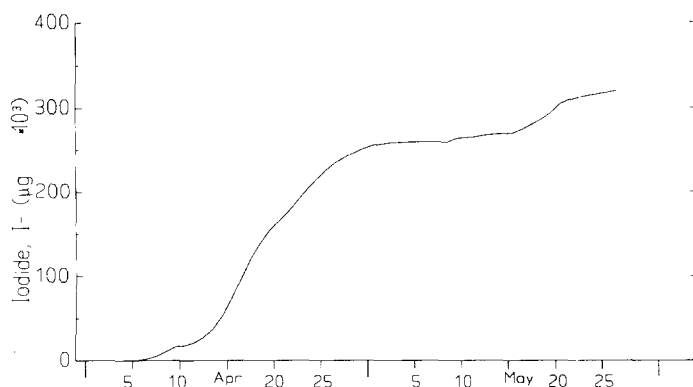


Fig. 6. Cumulative total of discharged tracer ($\times 10^3 \mu\text{g}$) detected at the V-notch during the spring melt 1 April–3 June 1987.

possible role when the matrix is unsaturated. Macropores can drain rainwater from the surface and not just as ordinary drainpipes which drain the soil body around the pipes. One objection to this conclusion is that macropores are usually obstructed and at a certain points sealed or terminated (Levy, 1987). But this dead end is of temporary nature and may therefore be a starting point for erosion if ruptured. Thus finer soil particles can be eroded leaving a coarser structure with higher permeability. Such erosion has been mentioned by Knutsson (1971) and Berry (1970; cited in Whipkey and Kirkby, 1978). By means of the stratigraphic investigation (Espeby, 1989) it is also concluded that several of the macropores of old root channels have long conducting flow paths, and some have their ending in the layer above the dense basal till. This till layer which has been subject to wave-washing has an assorted texture usually in the range of coarse sand, and thereby a high permeability (saturated hydraulic conductivity: 10^{-5} – 10^{-4} m s^{-1}). It is evident that macropores can play an important role in transmitting precipitation down to deeper parts of the soil body and thereby contributing to the chemical composition of the runoff water. This may occur even when almost the entire soil matrix is frozen. The consequence of this quick conductance through macropores is that the runoff water is not chemically buffered as it usually is when it passes through the soil matrix.

A hydrograph separation using ratios of electrical conductivity (Espeby, 1988) indicated that the melting component contributed between 43 and 81% of the total runoff during the highest runoff peak (25 April 1986), depending upon what concentration level of input water was assumed. These assumed concentrations ranged from 'pure' meltwater (snowcores) with an electrical conductivity of 1 mS m^{-1} to a 'contaminated' water that had been transported very quickly but shallowly with an electrical conductivity of 10 mS m^{-1} . The more pronounced acid surge at point 5 compared with points 3 and 4 during the last snowmelt indicates that some macrostructure had conducted meltwater to

deeper parts of the soil body. Boulders or other macrostructures in the upper 0.5 m of soil must have had an effect on the water conductivity in this case.

The accumulated amount of discharged I^- -tracer at the V-notch was very small compared with the amount expected to discharge under the saturated conditions that prevail during snowmelt. One possible reason for the small amount of discharged tracer was that some of the tracer applied to the area above point 6 may have leaked through, or under, the road embankment in the lower part of the catchment during the highest discharge peak. Another more reasonable explanation of the low amount of discharged tracer is that much of this tracer was retained in the soil matrix because of iodide attachment to positive iron-complexes and organic matter in the unsaturated zone. A much higher concentration of the applied tracer would have been more appropriate in order to suppress the iodide attachment in the matrix. Despite inability to permit the turnover time calculation, the experiment has given more support to the hypothesis of rapid water movement in macropores, that was derived by ordinary water chemistry analysis.

All these analyses have shown that the concept of piston flow is not always valid during the snowmelt period. This is partly because of the inhomogeneities in the soil structure and partly because of an inhomogeneous melting of the frost present in the ground. The three different methods used in this study have all indicated a need to interpret the results differently if the soil is frozen or not during time of sampling. When thawing releases the frost-bound water in the soil, the mechanism of water movement changes. During snowmelt, when the soil is frozen, the effect can lead to rapid preferential water movements in coarser parts of the matrix (although under unsaturated conditions). When the frost melts the matrix becomes saturated and traditional piston flow takes over. The large impact of the acid meltwater on the chemical composition in the spring (point 5) during the major snowmelt event in 1986 and no (or little) impact during rain after the frost release would be an example of this mechanism.

It is, however, important to notice the distinction between the samples from point A (the seepage area outside the catchment) and the samples taken in the drained catchment. The piston flow effect seems to be active at point A throughout the snowmelt but not in the drained catchment, as a result of a generally higher groundwater table throughout the year. The chemical analysis only indicated a near-surface transported water while the ^{18}O analysis from sample point A indicated that some displacement of old soil/groundwater was taking place during the highest runoff peak of the 1986 snowmelt.

Concluding remarks

Water from a macropore was almost similar to precipitation water, which indicates that little chemical reaction had occurred as a result of rapid runoff and limited contact with the soil.

Macrostructures that are present both in vertical and in horizontal directions in glacial till can promote water movement to deeper soil horizons along preferential pathways in contrast to diffuse flow.

This study has also shown that chemical analysis is a simple and cheap method to use in deriving the origin of waters where otherwise more expensive tracer methods would be required.

Iodide is not a good tracer under unsaturated conditions and if used it should be introduced beneath the soil B horizon.

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